

Operationalizing a Spacetime Hole: Traversal Cost, Defects, Repulsion, and Connectivity

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The phrase “a hole in spacetime” conflates four operationally distinct questions concerning curvature, force, traversal cost, and connectivity. We make the intuition of “less to traverse” precise through a reference-relative boundary kernel E_U : the difference in earliest reception time between a region and a reference filling with the same calibrated boundary. In a static spacetime, light probes the optical metric h_{ij}/N^2 , massive trajectories probe an energy-dependent Jacobi metric, and propulsive effort remains protocol dependent. A regularized surplus-angle cone has negative curvature and increased transverse optical cost but unchanged radial cost and no ordinary radial force, showing that curvature sign alone does not diagnose a shortcut. By contrast, a spherical thin shell with a negative Schwarzschild exterior contracts the optical metric and produces a Shapiro advance, demagnification, and composition-independent outward free fall. The shell requires exotic surface stress and presents an energy barrier to inward-moving massive particles. Its four-volume form nevertheless equals the flat expression in areal coordinates, so volume density is not the controlling traversal observable. Deleting a region while holding the metric cost fixed cannot produce a shortcut: an advance requires a lower path cost, an enlarged path set, or both. We conclude with coarse-grained and multispecies tests for a universal effective geometry and a six-item reporting checklist.

I. INTRODUCTION: OPERATIONAL QUESTIONS BEHIND THE METAPHOR

The phrase “a hole in spacetime” is evocative but underspecified. It may suggest missing microscopic constituents, reduced spatial volume, negative curvature, repulsive free fall, an early-arriving signal, an excised region, or a bridge between otherwise distant locations. None of these meanings implies the others automatically. The purpose of this paper is to separate the operational content of the metaphor from associations supplied by an external picture of the manifold.

An embedded observer instead works with coincidences of events, causal order, calibrated clocks and volumes, signal responses, and the motion of test bodies [1, 2]. The intuition pursued here concerns the time or effort needed to traverse a region. Its closest geometric realization is not a scalar density of spacetime but a cost assigned to directions and paths relative to a specified reference. For light in a static spacetime this cost is encoded exactly by the optical metric. For massive particles it also depends on energy. “Effort” is more protocol dependent still: free fall costs no proper acceleration, whereas hovering, prescribed transit time, and rocket fuel define different functionals.

It is therefore useful to ask four questions separately. Do neighboring paths spread? Are freely falling bodies pushed? What does an existing route cost? And which routes exist in the first place? Table I associates each question with its natural measurements.

The separation also resolves an ambiguity in the word *shortcut*. Earlier arrival is an outcome. Lower metric cost and changed connectivity are distinct mechanisms that can produce it.

The remainder of the paper is organized as follows. Section II defines the reference-relative boundary kernel. Sections III and IV formulate null and massive traversal costs, respectively. Sections V and VI separate angle defects from lapse-driven repulsion, and Sec. VII supplies an exact thin-shell realization. Section VIII proves that excision at fixed cost cannot create a shortcut. Sections IX and X state the conditions under which microscopic or coarse-grained data can support a universal effective geometry. Section XI compares the measurement protocols and relevant scales, and Sec. XII summarizes the operational conclusions.

II. BOUNDARY DATA AND A REFERENCE-RELATIVE KERNEL

An observer confined to spacetime has access to coincidences of events, causal order, clock readings, freely falling bodies, signal responses, and volume measurements. At resolution ϵ we summarize these data schematically as

$$(X_\epsilon, \preceq_\epsilon, \tau_\epsilon, \nu_\epsilon, \{\mathcal{R}_{\epsilon,a}\}), \quad (1)$$

where a labels probe species or channels. Under standard causality hypotheses, causal relations determine the conformal geometry, while calibrated volume data determine its remaining scale [3, 4]. At finite resolution it is safer, however, to state claims directly in terms of measured boundary data.

Let a world tube B of calibrated emitters, receivers, and clocks surround a region U . For a specified channel and protocol, let $\mathcal{T}_U(b_1, b_2)$ be the infimum of reception times over all admissible signals from b_1 to b_2 . Multiple images are included, and caustics are resolved by

Question	Operational observable	Geometric structure	Does not determine
Do paths spread?	circumference ratios, lensing	curvature, angle, holonomy	force or travel time
Are bodies pushed?	released-body acceleration, redshift	lapse gradient ∇N	topology
What does a route cost?	arrival-time kernel, energy threshold	optical or Jacobi metric	curvature sign
Which routes exist?	path classes, images, bridges	connectivity, topology	local force or cost

TABLE I. Four operational questions contained in the word “hole.” The entries are deliberately nonexclusive: for example the lapse contributes to both redshift and optical cost, but neither observable fixes the other sectors without additional information.

the protocol. Choose a specified reference filling U_0 , not necessarily flat, together with a fixed boundary isometry $\iota : B_0 \rightarrow B$ that identifies emitters, receivers, the induced boundary metric, clock calibration, and emission prescription. A flat reference is used only when it is compatible with these boundary data. The boundary time is proper time, $d\tau_B = N_B dt$; the same N_B is used in both fillings and may conventionally be normalized to unity. We then define

$$E_U(b_1, b_2) = c[\mathcal{T}_U(b_1, b_2) - \mathcal{T}_{U_0}(b_1, b_2)]. \quad (2)$$

Thus $E_U > 0$ is a delay, $E_U < 0$ an advance, and $E_U = 0$ neutrality in that channel. This kernel is the primary operational meaning of more or less to traverse. It does not require an arbitrary point-by-point identification of the two interiors. It is invariant under interior coordinate changes but depends, as every relative observable must, on the chosen pair (U_0, ι) .

III. NULL TRAVERSAL COST: THE OPTICAL METRIC

Consider a static spacetime

$$ds^2 = -N(\mathbf{x})^2 c^2 dt^2 + h_{ij}(\mathbf{x}) dx^i dx^j, \quad N > 0, \quad (3)$$

with the time normalization fixed at the comparison boundary. This fixing is essential: under $t' = \alpha t$ one has $N' = N/\alpha$ and $h_{ij}^{\text{opt}'} = \alpha^2 h_{ij}^{\text{opt}}$. Once the same boundary proper time has been chosen in the physical and reference fillings, the optical comparison is unambiguous. Along a null curve,

$$c dt = \frac{\sqrt{h_{ij} dx^i dx^j}}{N} = \sqrt{h_{ij}^{\text{opt}} dx^i dx^j}, \quad h_{ij}^{\text{opt}} = \frac{h_{ij}}{N^2}. \quad (4)$$

The spatial projections of light rays are stationary curves of this optical length, and the earliest admissible ray realizes its infimum under the stated boundary protocol [5, 6]:

$$\mathcal{T}_U(b_1, b_2) = \frac{1}{c} \inf_{\gamma: b_1 \rightarrow b_2} \int_{\gamma} \sqrt{h_{ij}^{\text{opt}} dx^i dx^j}. \quad (5)$$

Because $dt = d\tau_B/N_B$ at the receiver network, the infimum in Eq. (5) is directly the earliest boundary-clock

reading after the fixed normalization. Every local freely falling observer still measures the vacuum speed of light to be c . An advance is a global comparison of optical lengths and boundary clocks, not a local superluminal speed.

If a map between the reference interiors has additionally been chosen, a directional cost ratio can be defined by

$$\eta_{\text{dir}}(x, v) = \frac{\sqrt{h_{ij} v^i v^j}/N}{\sqrt{h_{ij}^{(0)} v^i v^j}/N_0}. \quad (6)$$

Then $\eta_{\text{dir}} < 1$ means less optical cost in direction v . A scalar geometric-mean proxy in three spatial dimensions is

$$\eta_{\text{vol}}^{\text{opt}} = \left(\frac{\det h^{\text{opt}}}{\det h_0^{\text{opt}}} \right)^{1/6}, \quad (7)$$

but it hides anisotropy: its value can be below unity even when some directions cost more. The full quadratic form, or ultimately the boundary kernel (2), is more informative.

In a weak, approximately isotropic field,

$$ds^2 \simeq - \left(1 + \frac{2\Phi}{c^2} \right) c^2 dt^2 + \left(1 - \frac{2\gamma\Phi}{c^2} \right) d\mathbf{x}^2, \quad (8)$$

so the directional optical cost relative to the coordinate reference is

$$\eta_{\text{dir}} \simeq 1 - \frac{(1 + \gamma)\Phi}{c^2}. \quad (9)$$

With the convention $\Phi = -GM/r$ for ordinary positive mass, this is a delay. A positive potential perturbation, as produced by the negative-mass toy model below, instead gives an advance. This is the limited sense in which a refractive-index analogy is useful; it should not be promoted to a fundamental medium without tests of dispersion and universality.

IV. MASSIVE PARTICLES: TIME, ACCESS, AND EFFORT

For a freely falling particle of rest mass m and conserved Killing energy E in (3), the spatial trajectory at

fixed E is governed by the Jacobi metric [7]

$$j_{ij}^{(E)} = \frac{E^2 - m^2 c^4 N^2}{c^2 N^2} h_{ij}. \quad (10)$$

It exists only where

$$E \geq mc^2 N. \quad (11)$$

Unlike the null optical metric, $j_{ij}^{(E)}$ is energy dependent. Different massive beams can therefore assign different effective traversal structures to the same spacetime even when all follow one spacetime metric.

Propulsive effort is a further observable, not another name for geodesic length. One possible protocol assigns a prescribed worldline Γ the integrated proper acceleration

$$\mathcal{A}[\Gamma] = \int_{\Gamma} \sqrt{a_{\mu} a^{\mu}} d\tau, \quad (12)$$

but fuel use also depends on the vehicle, exhaust velocity, payload, and speed profile. Free fall has $\mathcal{A} = 0$ whether the trip is long or short. Consequently “less time” and “less effort” should be reported as separate, protocol-labelled quantities.

V. ANGLE DEFECTS DO NOT MEASURE DEPLETION

The geometric theory of crystal defects provides a more specific continuum-mechanical realization: disclinations carry curvature, whereas dislocations carry torsion. In the Volterra construction, removing or inserting an angular sector produces the conical geometry considered here [8]. Remove a wedge from a disk and glue its edges, or insert a wedge and glue. Outside the core,

$$h_{\beta} = dr^2 + \beta^2 r^2 d\phi^2, \quad 0 \leq \phi < 2\pi. \quad (13)$$

For $\beta < 1$ the circumference $2\pi\beta r$ has a deficit; for $\beta > 1$ it has a surplus. The curvature concentrated at the apex has total weight

$$\int K dA = 2\pi(1 - \beta). \quad (14)$$

A smooth core of radius a is obtained from

$$h = dr^2 + b(r)^2 d\phi^2, \quad b(r) = r + (\beta - 1) \left[r - \frac{\sqrt{\pi}a}{2} \operatorname{erf}\left(\frac{r}{a}\right) \right]. \quad (15)$$

Here $b'(0) = 1$, $b'(\infty) = \beta$, and

$$K(r) = -\frac{b''(r)}{b(r)} < 0 \quad (r > 0, \beta > 1). \quad (16)$$

The integrated curvature remains $2\pi(1 - \beta)$ because $\int K dA = -2\pi[b'(\infty) - b'(0)]$; it is independent of the

regulator size a . The defect charge is therefore conserved while its curvature density is smeared. For $\beta \neq 1$ the geometry is asymptotically conical, so comparisons with Euclidean space must be made at a specified boundary and with an explicit reference subtraction.

The traversal interpretation now gives a sharp result. When $N = 1$, the optical metric equals (13). Radial cost is exactly dr , whereas angular cost is $\beta r d\phi$. A surplus defect has negative curvature but *more* transverse cost; a deficit has less transverse cost but positive concentrated curvature. Curvature sign is therefore not a density of what a ray must traverse.

In the static cylindrical extension

$$ds^2 = -N(r)^2 c^2 dt^2 + dr^2 + b(r)^2 d\phi^2 + dz^2, \quad (17)$$

an ideal cone with $N = 1$ is locally flat away from the axis. It has global holonomy and lensing-like effects, as for cosmic strings [9], but no ordinary local radial force. In $2+1$ Einstein gravity this distinction is exact: source-free regions are flat and localized sources act through global conical geometry [10].

VI. REPULSION REQUIRES A LAPSE GRADIENT

In the weak-field limit,

$$N^2 = 1 + \frac{2\Phi}{c^2}, \quad \ddot{x} \simeq -\nabla\Phi. \quad (18)$$

A smooth positive hump

$$\Phi(r) = \frac{\Phi_0 a}{\sqrt{r^2 + a^2}}, \quad \Phi_0 > 0, \quad (19)$$

gives

$$-\partial_r \Phi = \frac{\Phi_0 ar}{(r^2 + a^2)^{3/2}} > 0. \quad (20)$$

It simultaneously lowers weak-field optical cost through (9). The force and the advance arise from related but different combinations of metric components: slow free fall probes a lapse gradient, while null travel probes h_{ij}/N^2 .

Equations (17)–(19) are kinematic until their effective stress tensor

$$T_{\mu\nu}^{\text{eff}} = \frac{c^4}{8\pi G} G_{\mu\nu}[g] \quad (21)$$

has been computed and tested. Negative spatial curvature cannot replace this dynamical calculation. For complete asymptotically flat data satisfying the appropriate energy conditions, positive-mass results impose strong restrictions on isolated negative total mass [11]. A background-relative underdensity or a compensated construction is therefore more conservative than an isolated negative-mass object.

VII. EXACT REALIZATION: A REPULSIVE THIN SHELL

We now set $c = 1$. Take a flat interior with time T ,

$$ds_-^2 = -dT^2 + dr^2 + r^2 d\Omega^2, \quad r < a, \quad (22)$$

and a Schwarzschild exterior with time t ,

$$\begin{aligned} ds_+^2 &= -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2, \\ f(r) &= 1 - \frac{2GM}{r}, \quad r > a. \end{aligned} \quad (23)$$

Let $M = -|M|$, so

$$f(r) = 1 + \frac{2G|M|}{r} > 1. \quad (24)$$

The shell Σ is $r = a$ with proper time τ . Continuity of its induced metric requires

$$dT = d\tau, \quad dt = \frac{d\tau}{\sqrt{f(a)}}, \quad ds_\Sigma^2 = -d\tau^2 + a^2 d\Omega^2. \quad (25)$$

Choose the unit normal to point toward increasing r and use $K_{ab} = -n_\mu \nabla_a e^\mu_b$. The nonzero mixed components are

$$K^{-\tau}_\tau = 0, \quad K^{-\theta}_\theta = K^{-\phi}_\phi = \frac{1}{a}, \quad (26)$$

$$K^{+\tau}_\tau = \frac{GM}{a^2 \sqrt{f(a)}}, \quad K^{+\theta}_\theta = K^{+\phi}_\phi = \frac{\sqrt{f(a)}}{a}. \quad (27)$$

With

$$S^a_b = \frac{1}{8\pi G} ([K]\delta^a_b - [K^a_b]), \quad (28)$$

the Israel junction conditions [12, 13] give

$$\sigma \equiv -S^\tau_\tau = \frac{1}{4\pi G a} \left(1 - \sqrt{1 + \frac{2G|M|}{a}} \right) < 0, \quad (29)$$

and

$$p \equiv S^\theta_\theta = \frac{1}{8\pi G a} \left[\frac{1 + G|M|/a}{\sqrt{1 + 2G|M|/a}} - 1 \right] > 0. \quad (30)$$

The induced 2 + 1 surface stress tensor violates all of the standard pointwise surface energy conditions: $\sigma + p < 0$ violates the null condition, $\sigma < 0$ violates the weak and dominant conditions, and the strong condition also fails. The shell is therefore an effective exotic source, not a proposal for ordinary matter. These signs do not by themselves settle radial stability. The radial shell potential depends on the equation of state; in particular, without specifying the perturbative response $\delta p/\delta\sigma$, neither stability nor instability follows.

A. Force and massive-particle access

Static exterior observers have four-velocity $u = f^{-1/2}\partial_t$. Their acceleration is

$$a_r = \partial_r \ln \sqrt{f} = \frac{GM}{r^2 f}, \quad a^r = \frac{GM}{r^2}. \quad (31)$$

For $M < 0$, hovering requires inward proper acceleration; a released body moves outward. The effect is composition independent because all test bodies follow the same metric geodesics.

Timelike geodesics, written in terms of specific energy $\mathcal{E}$ and specific angular momentum ℓ , satisfy

$$\dot{r}^2 + V_{\text{eff}}(r) = \mathcal{E}^2, \quad V_{\text{eff}} = f(r) \left(1 + \frac{\ell^2}{r^2} \right). \quad (32)$$

This potential decreases monotonically toward its asymptotic value as r increases, so there are no circular timelike geodesics in the exterior. The access condition (11) also exposes the difference between time and effort: since $N = \sqrt{f} > 1$, a particle released from rest at infinity has $\mathcal{E} = 1$ (equivalently physical energy $E_{\text{phys}} = m$) and cannot penetrate to finite r . Inward traversal needs additional kinetic or propulsive energy even though the null route is optically shorter. Outward traversal, by contrast, is assisted by the field.

B. Optical contraction without four-volume depletion

For (23), the optical metric is

$$d\ell_{\text{opt}}^2 = \frac{dr^2}{f(r)} + \frac{r^2}{f(r)} d\Omega^2. \quad (33)$$

Every principal optical length at fixed areal radius is contracted relative to flat space. In particular a radial increment costs

$$d\ell_{\text{opt}} = \frac{dr}{f(r)} < dr. \quad (34)$$

This is a precise realization of “less to traverse.” Figure 1 contrasts this optical contraction with the surplus-cone cost and the massive-particle barrier.

It is not, however, a statement about the four-volume density. On a static slice,

$$dV_3 = \frac{r^2 \sin \theta}{\sqrt{f}} dr d\theta d\phi, \quad (35)$$

which is smaller than the flat value at the same areal radius. This comparison is geometric rather than a radial-coordinate convention: r is fixed by the area $4\pi r^2$ of each symmetry sphere. The spacetime volume form is

$$dV_4 = N dt dV_3 = r^2 \sin \theta dt dr d\theta d\phi, \quad (36)$$

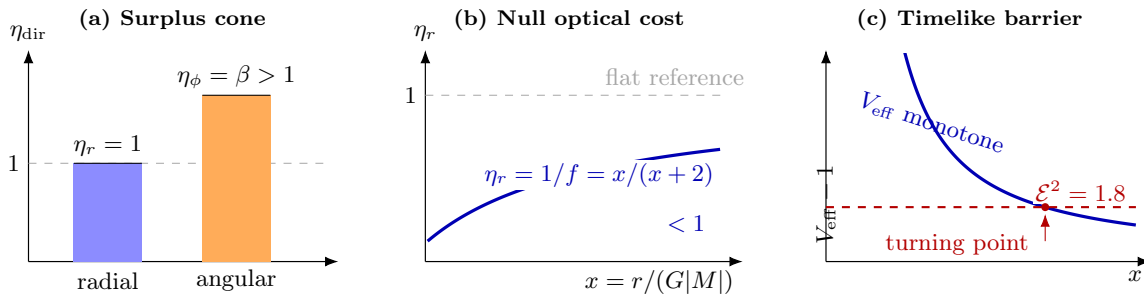


FIG. 1. Three operationally different effects. (a) For a surplus cone with $N = 1$, radial optical cost is unchanged while angular cost is enlarged: negative curvature does not mean less to traverse. (b) In the negative-mass Schwarzschild exterior, $x = r/(G|M|)$ and the radial null cost $\eta_r = 1/f = x/(x+2)$ lies below the flat reference. (c) For specific angular momentum $\ell/(G|M|) = 1$, the timelike effective potential decreases monotonically toward unity; finite specific energy produces a turning point rather than a circular orbit. The panels display intrinsic cost functions, not embedding diagrams.

exactly the flat expression in these geometrically fixed coordinates. The cancellation $N\sqrt{h} = r^2 \sin \theta$ is specific to the Schwarzschild-form metric in areal coordinates; it is not a general relation between repulsion and four-volume. The advance is controlled by h_{ij}/N^2 , not by $\sqrt{-g}$.

For impact parameter $b \gg G|M|$, the weak deflection is

$$\delta\phi = -\frac{4G|M|}{b} + O\left(\frac{G^2 M^2}{b^2}\right), \quad (37)$$

so rays are divergently deflected and a background source is demagnified in the weak-lensing regime. The leading Shapiro contribution also changes sign,

$$\Delta t = -2G|M| \ln\left(\frac{4r_e r_o}{b^2}\right) + O(M^2), \quad (38)$$

relative to a Minkowski filling with the same emitter and receiver areal radii, the same impact-parameter convention, and the same asymptotic time normalization. This gives $E_U < 0$ in the null channel. These are statements about the relevant optical congruence; full spacetime focusing remains governed by the null Raychaudhuri equation, with Ricci, Weyl, shear, and vorticity contributions [14].

With flat-space subtraction, the Brown–York energy of a round sphere is

$$E_{\text{BY}}(r) = \frac{r}{G} \left(1 - \sqrt{f(r)}\right) < 0, \quad (39)$$

consistent with quasilocal depletion relative to the reference [15]. Its numerical value is subtraction dependent and it is not a local energy density. The minimal construction also has negative Arnowitt–Deser–Misner (ADM) mass. A distant compensating positive shell can change the exterior asymptotics and total mass while leaving a negative quasilocal region, but its junctions and stability must be analyzed as part of the enlarged model.

The shell should therefore be read as an effective exotic source, not as an ordinary material proposal. Quantum inequalities constrain weighted negative energy densities for specified quantum fields and states [16]; they do

not automatically apply to an unspecified classical shell. Applying them here would require a concrete quantum source, sampling worldline, and sampling function.

VIII. DELETION IS NOT A SHORTCUT

The distinction between cost and connectivity follows from one variational observation. Write an earliest-arrival problem abstractly as

$$\mathcal{T}[\mathcal{C}, L] = \inf_{\gamma \in \mathcal{C}} \int_{\gamma} L, \quad (40)$$

where $\mathcal{C}$ is the set of admissible paths and L is their local cost. There are then two independent ways to obtain an advance:

1. lower L on some existing paths, as a metric or lapse change does;
2. enlarge $\mathcal{C}$, as a bridge or identification can do.

Deleting a closed region while keeping the surrounding geometry fixed does the opposite of the second operation: $\mathcal{C}' \subseteq \mathcal{C}$. The comparison assumes that L is held pointwise fixed on every surviving path. Therefore

$$\mathcal{T}[\mathcal{C}', L] \geq \mathcal{T}[\mathcal{C}, L]. \quad (41)$$

For an ordinary length space this is the familiar inequality $d_{Y \setminus H}(p, q) \geq d_Y(p, q)$. In a fixed spacetime, excision can cause detour, delay, scattering, disconnection, or no change, but not an earlier arrival. If removing matter changes the metric, the cost L has changed and the monotonicity statement no longer compares the same problem; this includes the physically common case in which matter removal changes h_{ij} or N .

Collapsing the boundary of an excised disk to a zero-cost point is not a surplus cone; it creates a pinched quotient. A physical bridge must have a nonzero throat geometry, a causal orientation, and a stress tensor. It must also be checked for non-Hausdorff behavior and

closed causal curves. Under the hypotheses of topological censorship, asymptotic flatness, global causal conditions, and averaged null energy restrict observable nontrivial topology [17]; these hypotheses should be stated rather than invoked as a slogan. Traversable wormhole models accordingly constitute a separate connectivity sector [18].

IX. IF “STUFF” IS MICROSCOPIC

A literal constituent interpretation requires an additional microscopic law. For example, causal-set models distinguish causal order from local finiteness and element count [19]. Even in such a model, fewer elements in a tube does not by itself imply earlier arrival: the order relations determine which chains exist, the count represents a volume-like quantity, and the dynamics must relate both to an effective metric. Thinning can reduce a count while also destroying routes. The continuum boundary kernel (2) is therefore a target observable that a microscopic model must reproduce, rather than something fixed by constituent density alone.

Kleinert’s second-gradient and nematic world-crystal models go further by proposing mechanisms through which Einstein-like long-range dynamics might emerge [20, 21]. These constructions require special constitutive or phase assumptions and therefore do not derive relativistic dynamics from defect kinematics alone. Moreover, a signed continuum defect density can discard neutral defect populations and their higher correlations [22].

The same warning applies to defect ensembles. A regularized two-dimensional collection of cone defects has the signed curvature measure

$$\Omega = \sum_i \alpha_i \delta_{p_i}, \quad \alpha_i = 2\pi(1 - \beta_i), \quad (42)$$

and a coarse-grained density in a window of area A_Λ ,

$$\bar{K}(\Lambda) = \frac{1}{A_\Lambda} \mathbb{E} \left[\sum_{p_i \in A_\Lambda} \alpha_i \right]. \quad (43)$$

This is meaningful as a two-dimensional or Regge-like curvature proxy. It is not a four-dimensional Ricci scalar and does not determine an optical cost. The traversal observable must be coarse-grained separately, for example

$$\bar{E}_a(\Lambda) = \mathbb{E}_{\mathcal{P}_\Lambda} [E_{U,a}(b_1, b_2)], \quad (44)$$

with the ensemble of boundary pairs $\mathcal{P}_\Lambda$ and probe channel a explicitly specified. In particular, $\bar{K}(\Lambda) = 0$ does not imply $\bar{E}_a(\Lambda) = 0$: optical length depends nonlinearly on the metric and on path minimization. The sign of $\bar{E}_a$ must be calculated for the specified ensemble rather than inferred from average defect charge.

Hausdorff and spectral dimensions can diagnose anomalous scaling, but they are not signed curvature

measures. In a smooth d -dimensional geometry the heat trace has the schematic small- t form

$$\text{Tr } e^{t\Delta} \sim (4\pi t)^{-d/2} (a_0 + a_1 t + a_2 t^2 + \dots), \quad (45)$$

where dimension appears in the leading exponent while curvature appears in subleading coefficients such as a_1 [23]. A running spectral exponent is useful evidence for a nonclassical regime, but its excess or deficit cannot be identified with the sign of scalar curvature without a specific model and limiting theorem.

X. COARSE-GRAINED UNIVERSALITY AND LOCAL FLATNESS

Three scales should remain distinct: probe wavelength λ_{probe} , laboratory size L_{lab} , and coarse-graining scale Λ_{cg} . They serve different purposes and need not obey one universal ordering. A fine wave can resolve and scatter from a defect even when a larger-scale average looks smooth. At a regular point p , a local inertial description requires, schematically,

$$L_{\text{lab}} \ll \text{dist}(p, H), \quad L_{\text{lab}}^2 \|\text{Riem}[g_{\text{eff}}](p)\| \ll 1, \quad (46)$$

where H denotes unresolved singular cores. A traversal curve approaching the reference value at short scale is not by itself an equivalence-principle theorem.

Before an effective response is called geometry, freely propagating species should share a causal cone. In a window $\epsilon \ll \lambda_{\text{probe}} \ll L_{\text{var}}$, where ϵ is a microscopic resolution scale and L_{var} is the shortest relevant background-variation scale, a useful high-frequency criterion is

$$P_a(x, k) = Z_a(x) g_{\text{eff}}^{\mu\nu}(x) k_\mu k_\nu, \quad Z_a(x) > 0, \quad (47)$$

for the principal symbol of every probe species a . Species-dependent cones, birefringence, or strong frequency dependence indicate an analogue medium rather than one universal spacetime metric. More concretely, after non-gravitational propagation delays have been removed, opposite signs of $E_{U,a}$ for two minimally coupled massless probes in the same high-frequency window rule out a single nondispersive effective metric for those probes. Outside that regime, differing signs need not be a geometric falsifier.

The four sectors in Table I can then be tested independently:

1. circumference and triangle data test angle defect and spatial curvature;
2. released test bodies and clock comparisons test lapse gradients;
3. boundary timing over directions, frequencies, and species tests traversal cost;
4. path multiplicity and causal routing test connectivity.

Agreement of these probes with one effective metric is a result to be demonstrated, not an assumption contained in the word “hole.”

For operational closure, any future claim of a space-time hole should therefore state six items: the reference filling and boundary identification, the probe channel and frequency window, the measured kernel $E_{U,a}$, the lapse-gradient response of released bodies and clocks, the angle or holonomy data, and the admissible connectivity. This checklist prevents one sector from being used as unreported evidence for another.

XI. DISCUSSION: MEASUREMENT PROTOCOLS AND SCALES

The word *observer* in this analysis denotes several measurement protocols, not four independent geometries. A freely falling local laboratory tests the local light cone and tidal corrections to inertial motion. Static observers compare clock rates and measure the acceleration required to remain at fixed position; Eq. (31) gives coordinate components of that acceleration, not a new scalar traversal cost. A boundary signal network measures the global kernel E_U by recording emission and reception times. The Brown–York quantity in Eq. (39) is different again: it is a quasilocal functional of a closed two-surface with a specified subtraction, not a local energy density. These protocols can be implemented by overlapping sets of observers, but their outputs answer different questions.

The examples deliberately cross, rather than occupy, single rows of Table I. The surplus cone changes angle and transverse optical cost while leaving the radial optical cost, the ordinary radial force, and the connectivity unchanged. The negative-mass shell changes the lapse, free fall, and null and timelike traversal costs while retaining the topology of the reference construction. Thus the four-sector table is a reporting discipline, not a one-to-one assignment of models or observers to rows.

The scale requirements are likewise logically separate. The window $\epsilon \ll \lambda_{\text{probe}} \ll L_{\text{var}}$ supports an effective continuum and a geometric-optics approximation, whereas Eq. (46) controls the size of a locally inertial laboratory. The coarse-graining scale Λ_{cg} must be specified by the microscopic model; there is no universal hierarchy that fixes its order relative to L_{lab} in every experiment. In particular, the contrast between the null advance and the timelike barrier of the shell is governed by the access condition (11), not by a wavelength hierarchy.

Finally, the relative observables do not all use interchangeable reference structures. The kernel E_U uses the filling and boundary identification (U_0, ι); the directional optical ratio requires a specified local comparison; the weak Shapiro advance uses the stated Minkowski comparator; and the Brown–York energy uses its own surface subtraction. Making each reference explicit is what makes the reported sign meaningful and reproducible.

XII. CONCLUSION

The intuition of less spacetime to traverse is viable when translated into a reference-relative path cost. In a static spacetime the translation is exact: light measures the optical metric h_{ij}/N^2 , and the boundary kernel E_U records delay or advance. Volume density, curvature, force, and connectivity provide other information but do not substitute for this cost.

The examples make the distinctions concrete. A surplus cone has negative curvature and enhanced transverse length but neither reduced radial cost nor a force. A repulsive negative-mass shell instead contracts the optical metric and advances light while presenting an energy barrier to inward-moving massive particles. It requires exotic surface stress, so it is an exact effective model rather than an ordinary source. Deleting a region with unchanged cost cannot create a shortcut; an advance requires a reduced metric cost or an enlarged path set. Any microscopic account of spacetime “stuff” must explain how its constituent density and connectivity produce these operational observables and why different probes see the same emergent geometry.

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